

EDUCATION

Georgia Institute of Technology, Atlanta, GA

Aug 2019 – Dec 2021

Master of Science in Electrical & Computer Engineering (Thesis)

GPA: 4/4

Coursework: Analog IC Design, Power IC Design, Analog Integrated Systems Design, Advanced VLSI Systems

Birla Institute of Technology and Science (BITS) - Pilani, Goa campus, India

Aug 2015 – May 2019

Bachelor of Engineering in Electrical & Electronics (Honors with Distinction)

CGPA: 9.38/10

TECHNICAL SKILLS

Core Skills – Power delivery, Voltage regulation, Switching DC-DC/buck converters, Multi-phase regulators

Tools/Programming – Cadence Virtuoso, LTspice, MATLAB, Spectre, ADICE, Simulink, Verilog-A, Python

WORK EXPERIENCE/INTERNSHIPS

Analog Devices, Inc., San Jose, USA

Aug 2021 – present (5+ years)

Senior Analog Design Engineer, High-Performance Power

- Chip lead for the **LT8340x-family**, switched-inductor DC-DC 42V ultra low-noise monolithic buck converter ICs.
- Led cross-functional closure across design, layout, Apps, test characterization, reliability, and product teams, including silicon debug, bench correlation, AEC-qualification/release readiness, datasheet/test-limit alignment and production.
- Built behavioral models to evaluate converter dynamics, load transients, loop stability, noise, and system-level tradeoffs.
- Closed loop noise optimization of *Silent-Switcher*® 3. Improved low-frequency noise by >30% over prior products.
- Developed coupled-inductor multiphase buck architecture targeting ripple cancellation, >200 kHz control-loop bandwidth, and μ V-level output switching ripple without any additional filter.

Georgia Institute of Technology, Atlanta, USA

Jan 2020 – July 2021

Graduate Research Assistant, Prof. Gabriel A. Rincón-Mora

- Analyzed and evaluated ADI's LT3922 LED Driver, prepared an eval plan and soldered board components.
- Designed Boost LED driver with Hysteretic current-mode control, aiming to study η_C in analog & PWM dimming.

Analog Devices, Inc., Santa Clara, USA

Aug 2020 – Dec 2020

Design Engineering Intern, LED Drivers Team

- Designed the LTspice model of **LT8386**, ADI's 60V boost LED Driver IC. Model published with product release.

Technion - Israel Institute of Technology, Haifa, Israel

June 2018 – Dec 2018

Research Intern, Prof. Shahar Kvatinsky

- Designed a Verilog-A based behavioral model for TowerJazz's single-poly Y-Flash (floating gate) multilevel memory.
- Developed a fully-integrated 4-bit trainable DAC with embedded flash device using an on-chip tuning mechanism.
- System specs: $f_S = 1$ MHz, $V_{DD} = 5$ V, $INL = 0.18$ LSB, $DNL = 0.05$ LSB, $ENOB \approx 3.4$ Bits.

RELEVANT PROJECTS

Two-stage Pipeline SAR ADC Design | Course: Analog Integrated Systems Design

Spring 2020

Prof. Shaolan Li, *Georgia Tech.*

- Transistor-level design of clock distribution, SAR logic, S/H, DAC switches and dynamic comparator.
- Designed V_{CM} -based differential CDAC to simplify SAR logic and minimize energy consumption.
- System specs: 6B+7B SAR, $f_S = 25$ MHz, $V_{DD} = 1.2$ V, $C_{DAC} = 10$ pF, $ENOB \approx 11.6$ Bits, $FOM_{Wal} = 37.3$ fJ/conv.

Type III Voltage-mode Asynchronous Boost Converter | Course: Power IC design

Fall 2019

Prof. Gabriel A. Rincón-Mora, *Georgia Tech.*

- Designed an asynchronous V-mode Boost switching DC-DC converter employing Type-III filter for stabilization.

High-Order Noise-Shaping SAR ADC Design and Optimization | Research project

Fall 2019

Prof. Shaolan Li, GAMMA Group, *Georgia Tech.*

- Investigated architecture choices of the NS-SAR ADC to realize power-efficient high-order noise shaping.
- Designed MATLAB-based behavioral models of NTF-optimized 2nd/3rd order EFB & CIFF Δ - Σ Modulators.

SELECTED PUBLICATIONS

- **V. Gupta** and G. A. Rincón-Mora, 'Dimming DC-DC LED Drivers,' *IECON*, Oct. 2021
- T-H. Wang, R. Wu, **V. Gupta** and S. Li, 'A 13.8-ENOB 0.4pF- C_{IN} 3rd-Order noise-shaping SAR in a single-amplifier EF-CIFF structure with fully dynamic hardware-reusing kT/C noise cancelation,' *IEEE ISSCC* and *IEEE JSSC*, 2021
- L. Danial, E. Pikhay, E. Herbelin, N. Wainstein, **V. Gupta et. al.**, 'Two-terminal floating-gate transistors with a low-power memristive operation mode for analogue neuromorphic computing,' *Nature Electronics*, Dec. 2019

AWARDS AND HONORS

- Received *ADI SPOT Award* for innovation and significant contribution by Analog Devices.
- Received *GT-ADI Fellowship* from Analog Devices, Inc. while pursuing M.S. at Georgia Tech.
- **1 of 3** students (from all 4 BITS campuses) offered the **BITSAA Graduate Application Scholarship**.